

Subject Description Form

Subject Code	AAE4112
Subject Title	Satellite System Engineering and Design
Credit Value	3
Level	4
Pre-requisite/ Co-requisite/ Exclusion	Pre-requisite: AAE2005 Electrics and Electronic for Aeronautical Engineering
Objectives	This subject will provide students with 1. Basic theory and concepts of spacecraft engineering. 2. Design concept, principle, theory, and methodology of satellite systems and subsystems. 3. Hands-on experience of conducting experiments, software simulations, and programming. 4. An opportunity to solve a practical engineering problem with a CubeSat design project.
Intended Learning Outcomes	Upon completion of the subject, students will be able to: a. Analyze practical engineering problems from a systematic and overall perspective. b. Identify and formulate the engineering problems related to the satellite topic with an understanding of engineering issues and constraints. c. Have preliminary capabilities for satellite (sub-)systems design and analysis and evaluate the design of the satellite based on the testing result and make necessary amendments to the design. d. Be able to use software tools to simulate the operations of the satellites.
Subject Synopsis/ Indicative Syllabus	Satellite mission – project management; system engineering; safety review process; safety requirements; environmental testing. Satellite subsystems – power control system; communication systems; command and data-handling system; structure system; mechanism system, thermal control system; attitude control system. Satellite mechanics – orbital science and mechanics; attitude mechanics; attitude control. CubeSat project – GMAT and MATLAB programming.

Teaching/Learning Methodology	Teaching is conducted through class lectures, case studies, and laboratory exercises. The basic knowledge, methodology, and theoretical analysis will be introduced in lectures. The understanding of how to design and test a satellite from a systematic engineering perspective is emphasized via case studies. Design methodology, analysis skills, satellite control techniques, and program methods are taught in laboratory sessions. Assignments and quizzes, laboratory exercises, projects, oral dissemination, and final reports are used to make up the coursework marks. <table><tr><th rowspan="2">Teaching/Learning Methodology</th><th colspan="4">Intended subject learning outcomes to be covered</th></tr><tr><th>a</th><th>b</th><th>c</th><th>d</th></tr><tr><td>1. Lectures</td><td>√</td><td>√</td><td>√</td><td>√</td></tr><tr><td>2. Case studies</td><td>√</td><td>√</td><td>√</td><td></td></tr><tr><td>3. Laboratory Exercises</td><td>√</td><td>√</td><td>√</td><td>√</td></tr></table>	Teaching/Learning Methodology	Intended subject learning outcomes to be covered				a	b	c	d	1. Lectures	√	√	√	√	2. Case studies	√	√	√		3. Laboratory Exercises	√	√	√	√										
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Assessment Methods in Alignment with Intended Learning Outcomes	<table><tr><th rowspan="2">Specific assessment methods/tasks</th><th rowspan="2">% weighting</th><th colspan="4">Intended subject learning outcomes to be assessed</th></tr><tr><th>a</th><th>b</th><th>c</th><th>d</th></tr><tr><td>1.Assignments/Quizzes</td><td>20%</td><td>√</td><td>√</td><td>√</td><td></td></tr><tr><td>2. Laboratory & Report</td><td>30%</td><td></td><td></td><td>√</td><td>√</td></tr><tr><td>3. Final Report & Presentation</td><td>50%</td><td>√</td><td>√</td><td>√</td><td>√</td></tr><tr><td>Total</td><td>100 %</td><td colspan="4"></td></tr></table> Explanation of the appropriateness of the assessment methods in assessing the intended learning outcomes: Overall assessment: 0.5 x continuous assessment + 0.5 x final report The continuous assessment (50%) is aimed at enhancing the students’ comprehension and assimilation of various topics of the syllabus via several laboratory teaching and laboratory report, numerical analysis, reading assignment. In particular, Laboratory & Report are used to assess the students’ capacities of self-study and problem-solving skills. In-class quizzes will be conducted to evaluate the students’ understanding of the teaching materials as well as attendance taking. Final report and presentation are to evaluate the student’s overall mastery level of the content of this subject.	Specific assessment methods/tasks	% weighting	Intended subject learning outcomes to be assessed				a	b	c	d	1.Assignments/Quizzes	20%	√	√	√		2. Laboratory & Report	30%			√	√	3. Final Report & Presentation	50%	√	√	√	√	Total	100 %				
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Student Study Effort Expected	Class contact:	
	▪ Lecture	27 Hrs.
	▪ Laboratory	9 Hrs.
	Other student study effort:	
	▪ Week 7 Reading week	3 Hrs.
	▪ Literature review / Scientific finding and analysis / report writing preparation	36 Hrs.
	▪ Self-study / preparation	36 Hrs.
	Total student study effort	111 Hrs.
Reading List and References	[1] Sebestyen, G. (2018). Low earth orbit satellite design. Springer. [2] C. Cappelletti, S. Battistini, and B. K. Malphrus, Cubesat handbook : from mission design to operations. London, England: Academic Press, 2021. [3] Q. Chen, Z. Liu, X. Zhang, and L. Zhu, Spacecraft power system technologies. Singapore: Springer, 2020. [4] Kaplan, E. and C. Hegarty, Understanding GPS: principles and applications. Artech House Publishers, Latest Edition. [5] V. A. Chobotov, Orbital mechanics, 3rd ed. Reston, Va: American Institute of Aeronautics and Astronautics, 2002. [6] C.-T. Chen, Linear system theory and design, 3rd ed. New York ;: Oxford University Press, 1999.	

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